
Age, climate and economic disparities drive the current state of global dam safety

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Supplementary Materials for

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Supplementary text

Data exploration of dam failure databases

In this investigation we collect information from the world register of large dams (WRD, ICOLD 2024) and their failure events (ICOLD 2019). These two databases contain information on the dams' owning country, typology, height, year of construction, amongst other variables. Fig. S1 depicts the characteristics of several variables for standing (black) and failed structures (blue), differentiating between concrete or embankment structures. Additionally, Fig. S1F shows the dam age at failure, clearly depicting a strong infant mortality phase. Fig.S1.C shows the ratio (in ‰) between dams that failed and the total dams built in the same year, in which we can appreciate a steady reduction of failure rate until the 60s.

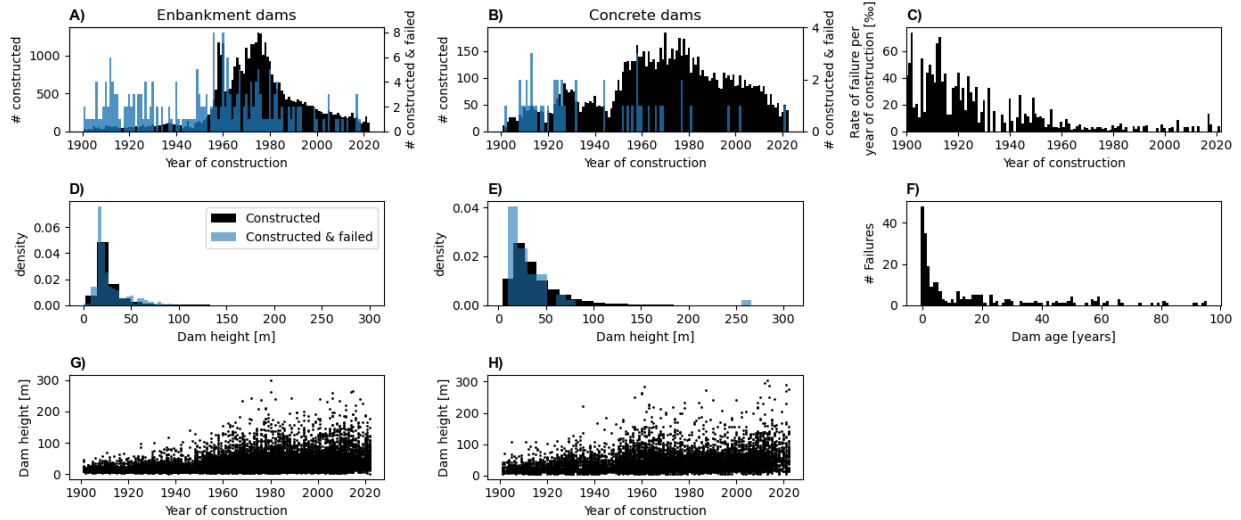


Fig. S1. Dam construction and failure year and height distribution for embankment (A, D, G) and concrete (B, E, H) dams. C) Estimated annual failure rate [%] for dams built after 1900, and F) number of failures vs. age of dam when failing.

Estimating the survival curve of large dams

Figure S2 shows non-parametric survival curves (Kaplan-Meier curves) for different partitions of the dam failure dataset. These curves discriminate between embankment and concrete type dams and, additionally, Fig. S2B, C and D show the effect in the estimated curve when selecting only dams built at different periods in history. This shows a clear infant mortality (higher probability of failure during the first years of operation), along with a relatively stable rate of failure with increasing age, showing that the dataset might not yet capture relevant ageing (acceleration of the failure probability). However, it shows the systematic improvement of dam safety in the second half of the 20th century.

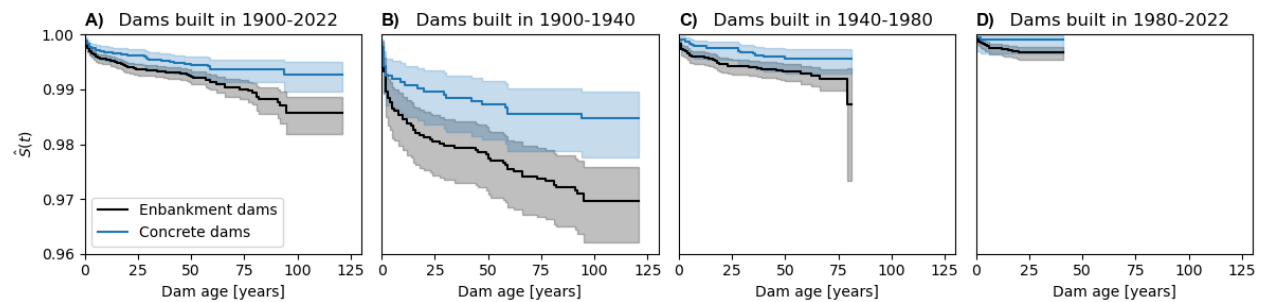


Fig. S2. Kaplan-Meier estimated survival curve ($\hat{S}(t)$) for embankment (black) and concrete (blue) dams when using: A) Dams built in the period 1900-2022, B) 1900-1940, C) 1940-1980 and D) 1980-2022. Solid lines represent the expected (mean), and the shaded bands the 95%CI of the survival curve, computed using the exponential Greenwood variance formula.

To test the effect of different covariates on the survivability of dams, we use two survival regression model families, namely a Cox proportional hazards model (which does not require assumptions on the underlying failure probability function), and a parameterized Weibull AFT model (which assumes a monotonic, Weibull distribution of failure rate). Additionally, we tested different functional representations of effects in the model. Table S1 shows the parameters adjusted at each model structure. The models were calibrated by the maximum likelihood method and a sensitivity analysis was performed to evaluate the robustness of the estimates to changes in the dataset. The performance of the models was evaluated through the metrics of concordance (higher the better) and Akaike Information Criterion (AIC, lower the better).

Additionally, a Weibull AFT model with non-linear effects for the year of construction (3rd-order spline) and height (quadratic model), and proportional effects for the rest of variables were chosen to perform forward predictions of failure probability. We tested the robustness of the fitted effects to biases in the failure database. For this, a sensitivity analysis was performed, evaluating the hazard ratio estimates when adding additional (0-50) synthetic failures randomly sampled, Fig S3 shows a low sensitivity of the estimated direction of effects to additional unreported observations or randomness of the failure process.

Table S1 Parameters fitted in each model, exponential of coefficients (hazard ratio), 95% CI, p-values (Wald two-sided significance test, null hypothesis: the parameter has no effect) and performance metrics for different survival model structures (CPH: cox proportional hazards, and Weibull AFT: Weibull accelerated failure time), uncertainties in the concordance metric were derived from 300 random samples with a random 70% data sampling. ED: Embankment dam, CD: concrete dam.

	CPH	CPH _{ICOLD} (only data until 2018)	CPH _{yh} (year-h)	CPH _{qh} (quadratic h effect)	CPH _{sy} (spline year of construction)	CPH _{fm}	Weibull AFT*
Parameter	Hazard ratio [95%CI] (p-value)	Hazard ratio [95%CI] (p-value)	Hazard ratio [95%CI] (p-value)	Hazard ratio [95%CI] (p-value)	Hazard ratio [95%CI] (p-value)	Hazard ratio [95%CI] (p-value)	exp(β)* [95%CI] (p-value)
GDP _{ppp} (k intUSD)	0.99 [0.984-0.997] (3.8E-3)	0.99 [0.984-0.996] (2.2E-3)	-	0.99 [0.98-0.99] (5E-3)	0.99 [0.98-0.99] (6E-3)	0.99 [0.981-0.997] (8E-3)	1.031 [1.007-1.055] (9E-3)
Height (m)	1.008 [1.003-1.013] (1.9E-3)	1.008 [1.003-1.014] (1.12E-3)	1.004 [0.999-1.01] (0.09)	1.018 [1.003-1.033] (0.01)	1.008 [1.002-1.013] (3E-3)	1.017 [1.002-1.032] (0.02)	0.93 [0.88-0.99] (0.02)
Height × Height (m ²)	-	-	-	0.99 [0.99-1.00] (0.20)	-	0.99 [0.99-1.00] (0.21)	1.0002 [1-1.00065] (0.21)
Dam type (ED:1, CD:0)	2.31 [1.65-3.23] (1E-6)	2.29 [1.62-3.21] (1.6E-6)	-	2.39 [1.66-3.31] (4.5E-7)	2.34 [1.67-3.29] (6.7E-7)	2.43 [1.67-3.33] (3E-7)	0.041 [0.01-0.15] (1.1E-6)
Year of construction	0.969 [0.965-0.977] (2.5E-34)	0.968 [0.963-0.973] (3.8E-37)	0.974 [0.96-0.98] (4.5E-27)	0.969 [0.964-0.974] (9E-35)	Spline 3 rd order (b1, b2, b3) [0.69, 0.01, 0.11] [0.15-3.05, 0.005-0.054, 0.03-0.40] (0.62, 8E-12, 5.4E-4)	Spline 3 rd order (b1, b2, b3) [0.63, 0.016, 0.010] [0.14-2.80 0.005-0.05 0.03-0.36] (0.55, 8E-12, 3.4E-4)	Spline 3 rd order [2.7, 5.4E5, 2.8E4] [0.01-6.1E2 5.2E4-5.7E8 25-3E5] (0.71, 6E-11, 9.3E-4)
b ₀	-	-	-	-	-	-	9.29E7 [4.9E6-1.74E9] (1.4E-34)
ρ	-	-	-	-	-	-	0.27 [0.24-0.3] (2.1E-92)
Metrics							
Concordance	0.70 [0.65-0.76]	0.71 [0.66-0.77]	0.67 [0.62-0.73]	0.70 [0.65-0.76]	0.71 [0.65-0.77]	0.71 [0.65-0.76]	0.71 [0.66-0.77]
Akaike Information Criterion (AIC)	4860.55	4686.47	4892.35	4860.26	4856.39	4856.25	4074.58

*The adjusted parameters of the accelerated failure time and Cox proportional hazard models have a different interpretation and should not be compared directly. A parameter higher than one in a CPH model implies an increase in hazard, whereas it represents a reduction in the AFT model.

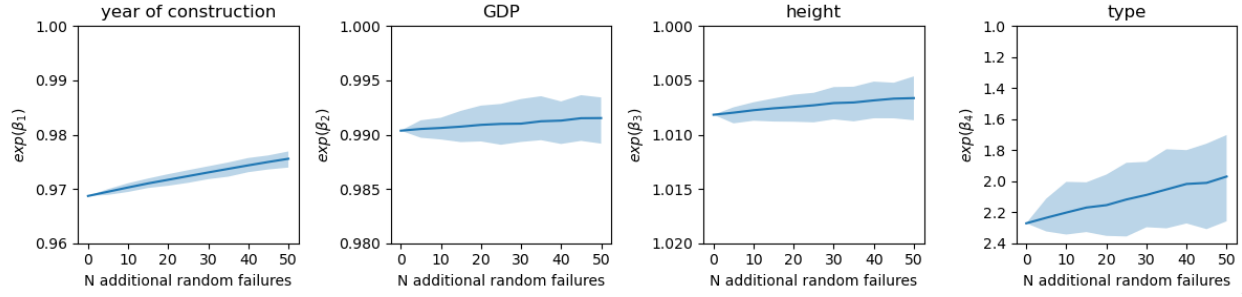


Fig. S3. Fitted coefficients for the four main variables when artificially adding a number ($N = [0, 50]$) additional failures randomly sampled over the dam population. Solid line: mean value, Shaded area: 95% CI over 500 samples.

Clustering high-risk dams

The Weibull AFT survival model (Table S1) was used to predict the probability of failure of all standing dams for the year 2023 (one year after the end of the dataset records) based on their prior history and characteristics, following Eq. 9 and Eq. 10 (Methods). We selected dams that exhibit a probability of failure above the strictest safety standards ($>1/10,000$) in 2023 and performed a clustering of these dams based on the similarity of their characteristics using a k-means clustering model (*sklearn* library, `cluster.KMeans`, Pedregosa et al., 2011). The clustering allows a more coherent description of the groups at higher risk. Fig. S5 shows the characteristics of the high-risk dams as distributed in four clusters of similar properties.

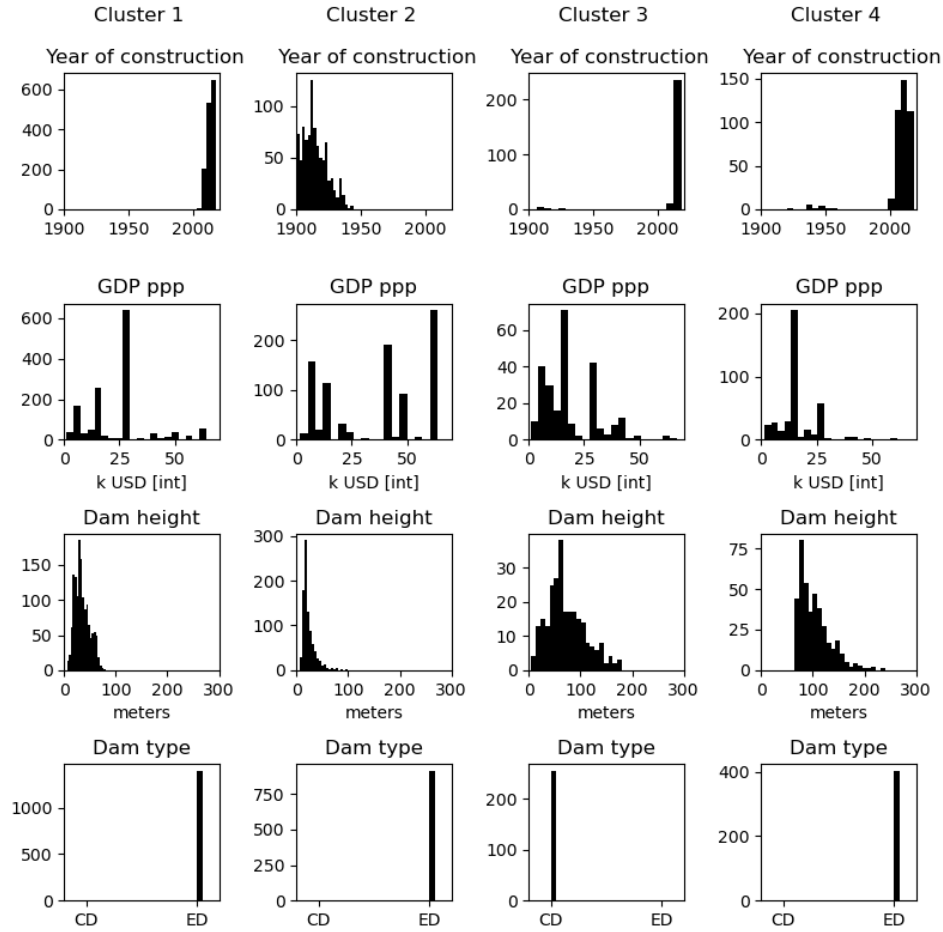


Fig. S4. Clustering of high-risk dams (for which probability of failure in 2023 is higher than 1/10,000) by means of a k-means model. ED: Embankment dam, CD: concrete dam.

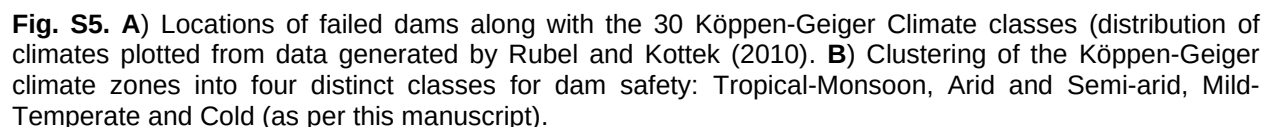
Hindcasting and predictions spatial validation

We performed a hindcast of past failures by country to assess the spatial representativity of the survival model predictions. Fig. 4C (manuscript) provides a list of 14 countries showing reported dam failures by ICOLD along with the predicted mean number of failures for the same period. Most countries' reported failures remain within the predictive confidence interval (CI), except for Japan showing a slightly safer past, and China, for which the number of reports ($N = 12$) represents a fraction of the historical failures predicted by our model (although informed by all other non-Chinese dam failure records).

Connections to climate

In this study we georeferenced the large dam failures reported by ICOLD (2019) bulletin. This was done manually by identifying the exact location of the dam when possible or the approximate location based on region and country. Using the dam failures coordinates, and the georeferenced WRD we sampled the climate zone for each dam location based on the Köppen climate zone global map produced by Rubel and Kottek (2010) updated for the year 2018. Additionally, the Köppen-Geiger climate classes were clustered in four types of relevance for dam safety, depicted in Table S2. Figure S5 shows the world Köppen climate classes, the clustered climate zones and the respective location of dam failures (1900-2024, excl. China).

Climate Cluster	Köppen-Geiger codes	Number of dam failures [1900-2022]	Description
Tropical and monsoon	'Af' : Tropical rainforest 'Am' : Tropical monsoon 'Cfa' : Humid subtropical 'Cwa' , 'Cwb' , 'Cwc' : Monsoon-influenced humid subtropical and oceanic climates 'Dwa' , 'Dwb' : Monsoon-influenced humid continental climates	101	Tropical and monsoon dominated climates. These are characterized by high annual precipitation and extremes.
<i>Arid and semiarid</i>	'As' , 'Aw' : Tropical savanna climates 'BSh' , 'BSk' : Semi-arid or steppe climates 'Csa' , 'Csc' : Mediterranean climates with dry summers 'Dsa' : Continental Mediterranean climate, hot summer 'BWh' , 'BWk' : Desert climates	52	Climates subjected to prolonged dry periods and water stress, coupled with highly-concentrated precipitation events.
<i>Mild-temperate</i>	'Cfb' , 'Cfc' : Oceanic climates with mild temperatures year-round. 'Csb' : Mediterranean climate with mild, wet winters and warm, dry summers. 'Dsb' , 'Dsc' , 'Dsd' : Continental Mediterranean climates 'Dfa' , 'Dfb' : Humid continental climates	84	Climates subjected to prolonged dry periods and water stress, coupled with highly-concentrated precipitation events.
<i>Tundra and cold</i>	'Dfc' , 'Dfd' , 'Dwc' , 'Dwd' : Subarctic or boreal climates 'EF' : Ice cap climate 'ET' : Tundra climate	6	Tundra and subarctic climates, characterized by heavy presence of ice and freeze-thaw cycles.



The role of extreme and unusual floods in past dam breaks

Data of ICOLD (2019) considers two types of large floods within the categorization “incident context”: unusual floods (large but below design flood) and extreme floods (above design floods). With the decrease in failures around the past century's end, the number of failures where a flood was important also decreased. It is also observed that the role of extreme floods (above design) decreased (period: 1950-2000), with unusual floods becoming a more usual cause of the incident. This could be linked to more stringent design considerations, requiring higher return periods for design, potentially established after dam incidents of the pre-80s. Fig S6A shows the ratio of overtopping (normally caused by rain-runoff processes) to other types of failures and the total number of failures where flooding was relevant (Fig S6B). The evolution of these ratios and failures per decade is depicted, number of reported failures in the 2020s decade are shown as per September 2024, and a linearly extrapolated value is shown as shadowed bars. Fig S7 also shows the evolution of the construction activity (by decade) per climate zone, along with the estimated failure rate and failure context.

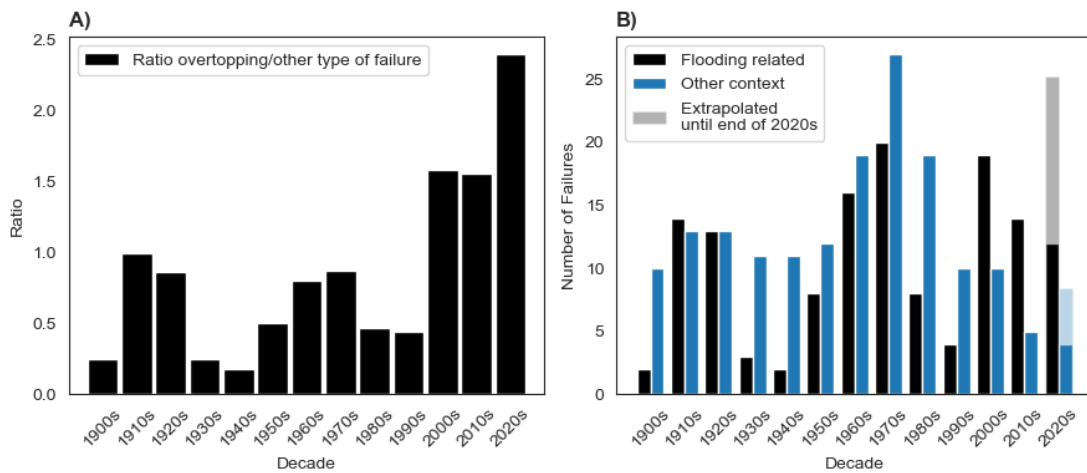


Fig. S6. Classification of the number of failures occurring per decade (according to year of failure), depending on type: **A)** ratio of overtopping related failures versus any other failure, and context: **B)** number of failures in which a flood was involved (black) versus non-flood related (blue), the decade 2020 is shown as per Sep-2024, and extrapolated to the full decade based on the present rates (semi-transparent bar).

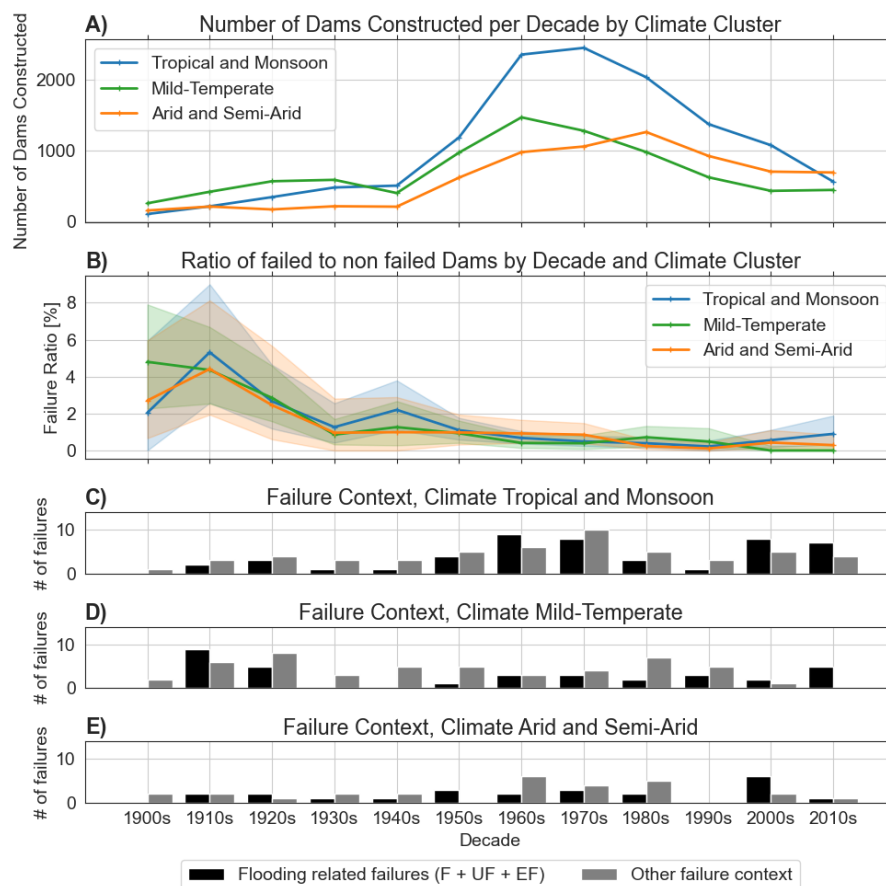


Fig. S7. A) Dams built per climate region by decade, **B)** Failure ratio of dams built per decade, with mean estimates (solid) and 95%CI (shaded). **C, D, E)** Number of failures per decade and per climate region by their failure context (related to a flood event or other causes, UF: unusual flood, EF: extreme flood, F: other flood, or other context, as identified in the ICOLDs Incident Database).

References

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